Toward Cross-Domain Class-Incremental Remote Sensing Scene Classification

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***Abstract*— Class-incremental (CI) learning has recently received extensive research interest in remote sensing scene classification (CI-RSSC). The existing CI-RSSC methods' superior performance seriously relies on old (base classes) and new classes (incremental classes) sampled independently from an identical distribution (dataset). In real-world RSSC scenarios, there exist significant distribution shifts between old and new classes, leading to the existing CI-RSSC methods being unable to adjust flawlessly to these new classes. In this article, we propose a novel cross-domain (CD) CI-RSSC framework to solve the above-mentioned problems, termed CDCI-RSSC. Specifically, a modular sharing-based dynamic extension module is first designed, which only updates specialized modules to extract new class feature embeddings for reducing memory footprint. Then, an effective dynamic alignment guided domain adaptive module (DAM) is further proposed to calculate the dynamic weights of each sample in various fields, which can minimize distribution shifts between source and target domains. Finally, a foreground enhancement module (FEM) is introduced to alleviate the issue of complex background interference in RSSC by increasing the weight of critical regions. Compared with the existing CI-RSSC and CD-RSSC, our proposed CDCI-RSSC framework surmounts the challenge of handling the distribution shifts between source (base session) and target domains (incremental session) while alleviating the limitations of continuous learning of new classes. Extensive experiments on three CDCI scenarios show that the CDCI-RSSC model achieves significant performance improvements in comparison to existing CI-RSSC and CD-RSSC methods.**

***Index Terms*— Class-incremental (CI) learning, domain adaptation (DA), foreground enhancement, remote sensing scene classification (RSSC).**

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I. Introduction

REMOTE sensing scene classification (RSSC) technologies can obtain high-level semantic information of the whole image by analyzing the various characteristics and properties of the remote sensing images, so as to realize the classification of different ground object types [1]. Compared with pixel-level classification, RSSC focuses more on the features and semantic information of the overall scene, which can better explain the complex ground object information of remote sensing data [2]. Over the past several decades, RSSC has been deeply applied in resource management, environmental monitoring, land use, and other fields [3].

In recent years, deep neural networks have garnered attention for their powerful capability to automatically extract high-level semantic information [4], and gradually applied to the field of RSSC [5]. For instance, Tong et al. [6] introduced a CAD network that enhances the extraction of multiscale features and vital characteristics to boost the accuracy of RSSC. Li et al. [7] put forward an ORRCNN method, which infers scene information using relationships to objects. Although these methods have made some progress in RSSC, they usually rely on retraining the entire model to accommodate emerging classes of features in different scenarios, imposing significant limitations for many practical applications [8].

To overcome the limitations of continuous learning of new classes, an effective class-incremental (CI) learning framework is proposed. The CI learning method allows classifying all categories without retraining the entire model and constantly enhances the generalization ability of the model in new classes [9]. The integration of CI learning into remote sensing has recently opened up new possibilities for RSSC, termed CI-RSSC. For instance, Liu et al. [10] introduced a CI learning framework based on open sets, which is used for RSSC in open and dynamic environments. Ye et al. [11] proposed an ECA-MSDWNet framework, which enhanced the model's ability to extract key information in complex environments and improved its task adaptability. Wang et al. [12] designed a G-MFCN model to improve the discriminability of new class features by enhancing the pretrained backbone's robustness. Although these proposed methods enhance the performance

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of CI-RSSC, their superior performance depends on the new and old classes being independently sampled from the same distribution, which is shown in Fig. 1(a).

Nevertheless, in the actual RSSC scenario, significant distributional shifts exist between new and old classes owing to differences in sensor types, illumination conditions, and shooting angles [13], [14]. These large shifts decrease the ability of the trained model to generalize well to new data. To solve this issue, some researchers have delved deeper into cross-domain (CD) learning in RSSC (CD-RSSC) [15]. For instance, Zhu et al. [16] designed a novel AMRAN method, which concurrently aligned both marginal and conditional distributions and further enhanced the model's adaptability. Yang et al. [17] proposed domain adaptation (DA) as one of the transfer learning methods, which solved the problem of distribution shifts by alignment of label-rich data (i.e., source domain) and label-sparse data (i.e., target domain). Although these DA methods address the domain gap issue across various datasets in RSSC, they assume that the categories between the two domains are consistent and can only classify classes present in the source domain, as shown in Fig. 1(b).

To overcome the shortcomings of existing CD-RSSC models in continuously learning new categories, as well as the shortcomings of CI-RSSC models in handling CD distribution shifts, a novel CD CI-RSSC framework, termed CDCI-RSSC, is proposed. Our proposed method is designed based on practical application requirements, wherein the target domain encompasses not only the existing categories in the source domain but also incorporates newly added categories. By integrating DA strategies into the incremental learning process, we achieve precise recognition of known categories and continuous learning of unknown categories under various data distribution conditions, as shown in Fig. 1(c).

To address the continuous emergence of new categories in CDCI learning, we design a modular sharing-based dynamic extension module, which efficiently extracts features of new classes while avoiding the substantial memory consumption associated with traditional dynamic expansion methods. Moreover, considering the potential distribution shifts between source and target domains, we incorporate DA techniques into the base learning process. The dynamic alignment and multiscale modules are leveraged to extract features at various levels and scales, thereby facilitating more effective alignment between features. Finally, a foreground enhancement module (FEM) is further designed to address the issue of complex background interference in RSSC by increasing the weight of key regions. The CDCI method significantly enhances the practical applicability of models in RSSC, effectively mitigating the challenges of distribution shift across different RS domains and overcoming the limitations of continual learning for new classes. Furthermore, the CDCI framework employs more sophisticated feature extraction techniques, improving the model's generalization capability and achieving excellent results across three different CDCI scenarios.

The main contributions are presented as follows.

1) We propose a simple and effective CDCI framework for RSSC, which can effectively address distribution shifts between different domains while overcoming the limitations of continuous learning for new classes. By investigating and highlighting the importance of CDCI tasks in remote sensing, we provide an innovative approach and perspective for RSSC.
2) We design a modular sharing-based dynamic extension method that updates only specialized modules to effectively reduce memory footprint and enhance feature extraction for new classes.
3) An effective dynamic alignment guided domain adaptive module (DAM) is proposed, which aims to obtain more comprehensive features and realize the targeted distribution alignment. Furthermore, an FEM is designed to mitigate negative transfer by emphasizing the weights of critical regions, which reduces the impact of background interference.

II. Related Work

In this section, some related works about CI learning and CD learning will be introduced in detail.

A. CI Learning

CI learning represents an advanced machine learning task that intends to achieve continuous learning in new categories without retraining the whole model [18]. The core challenge of this task is to avoid catastrophic forgetting while absorbing new information. Recently, this issue has received significant interest from numerous researchers. LWF stands out as a classical method in CI learning fine-tuning the new model through knowledge distillation technology to retain the knowledge from the previous model [19]. Zhang et al. [20] separated the feature extractor from the classifier, focusing solely on updating the classifier during the incremental phase. Hu et al. [21] introduced a method to relieve catastrophic forgetting by establishing causal routing between new and old data. Yu et al. [22] developed a dual knowledge distillation method, which can effectively prevent knowledge forgetting during the new task training.

In recent years, with the continuous maturity of CI learning methods, more and more researchers have begun to apply them to the field of remote sensing. Compared with traditional public datasets, remote sensing images have more complex backgrounds and often face the challenge of high Interclass similarity and large intraclass differences. To overcome these difficulties, a series of innovative approaches have been emerged. Ren et al. [23] introduced an LSAW method that addresses category imbalance and utilizes taxonomic information from errors in RSSC. Zhu et al. [8] developed a CI learning method for continuous prototype calibration that solves the challenge in fine-grained RSSC by calibrating interclass and intraclass prototypes. Wang et al. [12] put forward a G-MFCN framework for RSSC, aiming to bolster the resilience of the feature extraction process, which enhanced the capability to effectively extract features from newly introduced sample classes. Ye et al. [11] proposed an ECA-MSDWNet approach to strengthen the model's ability to extract information from intricate remote sensing environments.

Although these methods have made pleased progress, the effectiveness of current CI-RSSC methods depends on both

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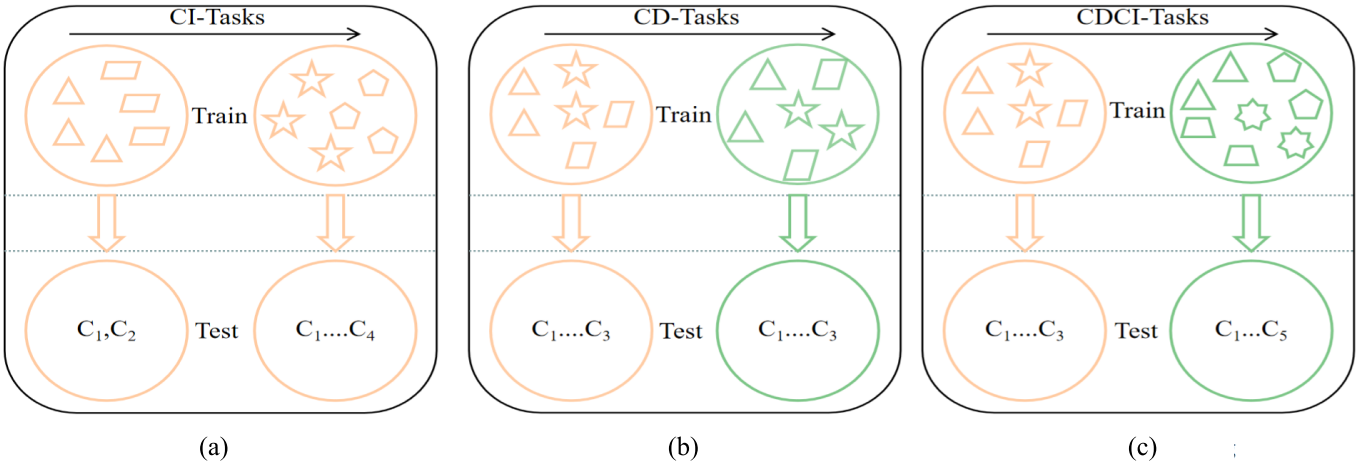


Fig. 1. Different experiment settings on (a) CI learning, (b) CD learning, and (c) CD CI learning methods. Colors indicate different domains, and shapes indicate different categories.

old and new classes being independently sampled from the same distribution. However, in the actual RSSC scenario, these methods cannot adapt flawlessly to new categories when there exist significant distribution shifts between old and new categories. Consequently, there is significant potential for the future integration of DA methods in CI-RSSC to address this challenge.

B. CD Learning

CD learning aims to address the issue of distribution shifts between source and target domains for enhancing knowledge transfer ability. Recently, researchers have explored CD learning and tried to improve the model's generalization ability across various distributed datasets. Long et al. [24] used a conditional domain adversarial network (CDAN) to effectively align the distribution of diverse domains by a multilinear conditional strategy. Pan et al. [25] put forward a transferable prototype network for classification by reconstructing the prototype distance to each class. Wei et al. [26] used a meta-alignment approach to minimize conflicts in optimization between tasks of domain alignment. Huang et al. [27] introduced a theory-based distributed adaptive method (DAET) to enhance the model's classification accuracy by integrating supplementary information from different domains.

The aforementioned CD learning methods address the problem of data distribution shifts between different domains but still face a series of challenges in practical application. First, in feature alignment, some methods tend to focus solely on alignment at a single level, overlooking the importance of aligning features across various scales. Second, they usually perform feature alignment from the overall level, without fully considering the influence of the relative importance between different samples. In particular, in the tasks of RSSC with complex backgrounds and blurred ground boundaries, these methods frequently struggle to achieve optimal results. To further address the problem of CD-RSSC, many DA techniques have been developed [28].

For instance, Yang et al. [17] considered the impact of each sample from various domains on the neural network's weights to better achieve distribution alignment by calculating dynamic weights in different domains. Huang et al. [29] proposed a TAADA network tailored for the characteristics of RS images. This network generates features that are both discriminative and invariant to domain changes by incorporating DA into the adversarial learning process. Zheng et al. [30] utilized a task-specific classifier framework to minimize the difference between two different classifiers, which generates more discriminative feature representations for CD-RSSC. Huang et al. [31] introduced a novel adversarial-learning-based DA network, which can be effectively applied to cross-scene classification tasks and significantly enhances the classification performance in the target domain.

Existing CD-RSSC methods require the same categories in the source and target domains. And even though the methods toward open datasets are able to distinguish between known and unknown classes, they still cannot support continual learning for newly emerged classes [32], [33]. Consequently, it is imperative to design a CDCI-RSSC framework that adapts to regional characteristics, robustly handles different distribution shifts, and consistently learns new categories.

III. Method

The existing CI-RSSC methods' great success highly relies on both old and new classes sampled independently from an identical distribution. In the actual RSSC scenarios, there exist significant distribution shifts between old and new categories, leading to the existing methods being unable to adjust flawlessly to new classes. In this article, we effectively combine CI learning with CD learning to propose a CDCI-RSSC framework, which effectively enhances the model's generalization capability. The model's overall structure is depicted in Fig. 2. First, CDCI only extends specialized modules to reduce memory footprint by dynamic extension based on module sharing. Then, an FEM is further proposed to improve the critical region weight and reduce background interference. Finally, a DAM is designed to extract features from various scales and levels and achieve dynamic alignment.

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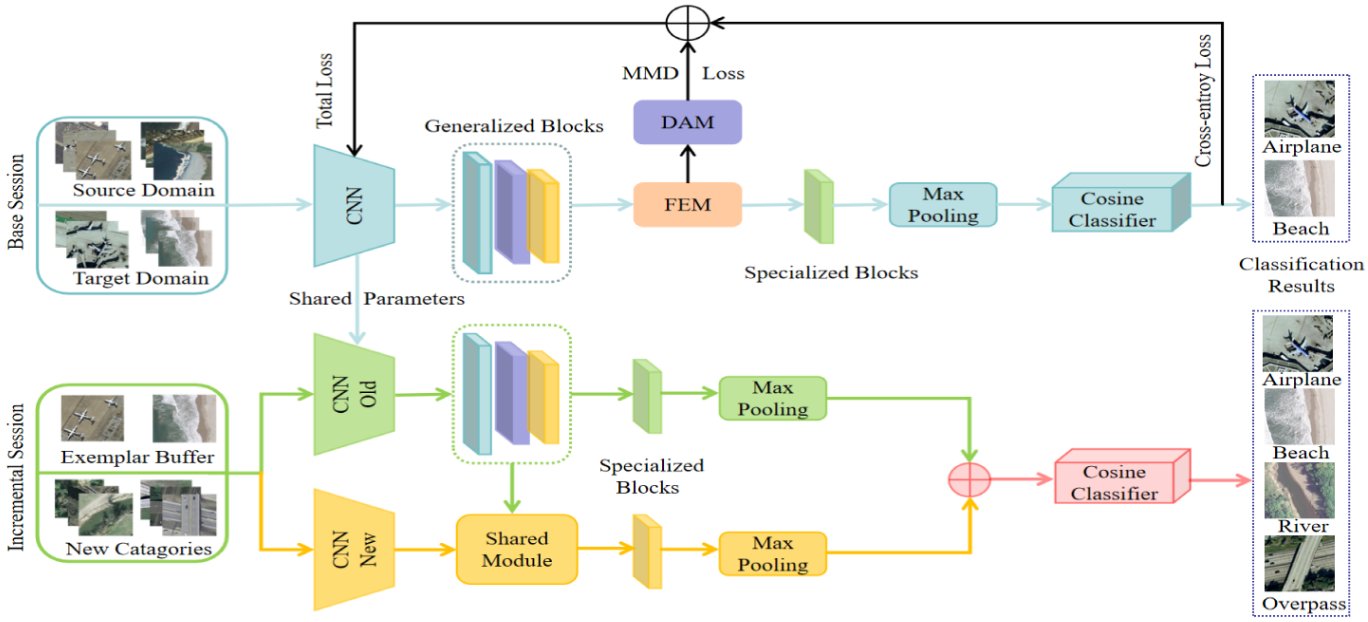


Fig. 2. Basic framework of the proposed CDCI-RSSC method. FEM represents the foreground enhancement module, DAM indicates the domain adaptive module, CNN is the convolutional neural network, MMD means maximum mean discrepancy, and $\oplus$ is the element-wise sum.

A. Problem Formulation

The CDCI method aims to tackle distribution shifts of various domains, while continuously learning new categories. In this article, D^S and D^T, respectively, represent the source and target domain datasets. The D^S data is used as the training set, and the label space corresponding to D^S is represented by C^S. D^T is divided into D^T_{new} (unknown categories) and D^T_{old} (known categories). C^T_{old} is the label space with unknown label information in the target domain D^T_{old}. D^T_{old} and D^S, respectively, share identical features and label spaces. For D^T_{new}, the training sets for various incremental learning phases are denoted by $\{D^{T1}_{\text{train}}, D^{T2}_{\text{train}}, \ldots, D^{Ti}_{\text{train}}\}$, where each dataset D^{Ti}_{train}has an associated label space represented by $C^{Ti}_{\text{train}}(i \geq 1)$. Here, there exist no overlapping classes across various datasets, $\forall i, j$ and $i \neq j$, $C^{Ti}_{\text{train}} \cap C^{Tj}_{\text{train}} = \emptyset$. For CDCI, the structure of the feature extraction network is trained using the training data from the initial session D^S. Generally, the dataset in the first session D^S has sufficient data for training, referred to as the base training set. Then, the pretrained network is employed to test the data in D^T_{old} and adapted to the new classes in incremental sessions $D^{Ti}_{\text{train}}(i \geq 1)$. In the incremental evaluation phase, $\{D^{T1}_{\text{test}}, D^{T2}_{\text{test}}, \ldots, D^{Ti}_{\text{test}}\}$ is applied to present the testing set for all tasks. For instance, in the AID $\rightarrow$ UCM scenario, the overlapping nine classes from both datasets are used as base classes. We utilize all data from the nine classes in AID as the training set and all data from the nine classes in UCM as the test set. After completing CD testing, the remaining categories in the UCM dataset are further served as new classes for incremental learning.

B. Modular Sharing-Based Dynamic Extension Method

CI learning intends to continually learn new categories while retaining knowledge of previously learned categories. Example-based methods preserve the memory of previous categories by replaying historical data. However, they are constrained by limited representational capabilities, and a monolithic model structure often fails to adapt adeptly to the dynamic characteristics of emerging categories. To overcome this issue, the DER [34] enriched the original model by adding new modules focused on specific tasks and categories, which helps to reveal key elements that the original model may have overlooked. The model is optimized with cross-entropy loss

$$L(x, y) = \sum_{k=1}^{|Y_b|} -I(y = k) \cdot \log S_k(W_{\text{new}}[\bar{\phi}_{\text{old}}(x), \phi_{\text{new}}(x)]) \tag{1}$$

where Y_b stands for all the seen classes and $S_k(\cdot)$ signifies the probability of the kth class following the softmax operation. W_{new} means the weight parameter of the new linear layer, $\bar{\phi}_{\text{old}}(x)$ is the frozen old feature embedding, and $\phi_{\text{new}}(x)$ corresponds to a new feature embedding.

However, as new categories continuously emerge, endlessly adding new modules will inevitably lead to a continuous increase in model parameters and memory usage. Moreover, in the scenario of CI learning, features at different levels have varying degrees of importance: shallow features tend to capture more general representations, while deeper features are more specialized. Based on this observation, a modular sharing-based dynamic expansion approach is designed, incorporating shallow networks as universal modules (specifically, the first three convolutional blocks of ResNet-18) and deep networks as specialized modules (the last convolutional block of ResNet-18). In each incremental learning stage, by sharing the general blocks, we only expand the specialized block to extract feature embeddings of new categories, which efficiently

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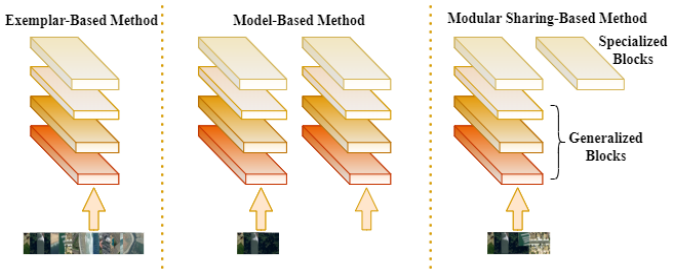


Fig. 3. Overview of three typical methods. (Left) Exemplar-based methods train a single model. (Middle) Model-based methods train a new model per new task. (Right) The modular sharing-based method trains a new specialized block per new task.

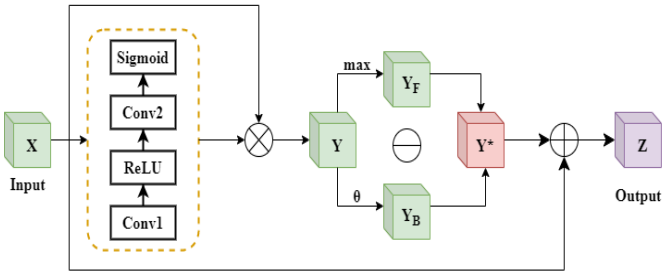


Fig. 4. Structure of the FEM. The refined foreground feature map is derived by subtracting the background feature map from the foreground, allowing for a heightened focus on object localization.

reduces memory usage. The loss function is described into

$$L(x, y) = \sum_{k=1}^{|Y_b|} -I(y = k) \log\Big(S_k\Big(W_{\text{new}}[\phi_{\text{sold}}(\phi_g(x)), \phi_{\text{snew}}(\phi_g(x))]\Big)\Big) \quad (2)$$

where $\phi_g(\cdot)$ represents the generalized block and $\phi_s(\cdot)$ denotes the specialized block.

Our innovative approach significantly optimizes memory management in models through module-sharing-based dynamic expansion techniques. Specifically, it achieves efficiency in memory usage by sharing common modules and customizing only the final module for new tasks. This saved memory reduces the overall budget requirements and provides the model with valuable space to store more old sample exemplars. These additional sample exemplars are essential for optimizing the model's training process, with effects that even surpass the strategy of adding common modules. Through this method, we improve the efficiency of the model and further enhance the adaptability for new categories. In Fig. 3, the difference between the method of dynamic expansion based on module sharing and other methods is shown.

C. Foreground Enhancement Module

In the CD learning of RSSC, the accuracy of image classification is significantly affected by background interference, leading to negative transmission when the model excessively focuses on background information. A feature map produced by a feature extractor typically has multiple channels. Specifically, different convolutional kernels capture distinct response regions within the input image. The information in each channel corresponds to a specific area of the image, which may be a background or an object. In the input image depicting rivers, certain channels exhibit a strong response to river features, capturing detailed river information, while others contain forests, buildings, and other background areas with low response. Consequently, the different channels contribute differently to the classification. In this article, the FEM is designed to filter the background information with low response. The FEM can alleviate background interference by increasing the weight of key regions and paying more attention to maintaining consistent features in all domains, so as to reduce the impact of interdomain differences. The structure is shown in Fig. 4, with detailed descriptions provided below.

First, we represent the output going through the shared model as $\mathbf{X} \in \mathbb{R}^{c\times h\times w}$, and then redefine the extracted features as $\mathbf{Y}^r \in \mathbb{R}^{c\times h\times w}$ through an attention mechanism, thereby allowing the target object regions to obtain more significant responses compared to irrelevant locations. Then, the maximum value of the global response is calculated as the most foreground feature map, which is shown as

$$Y_F = \max_{m,n} Y^r_{i+m,j+n} \quad (3)$$

where (i, j) is the position in the output tensor and (m, n) denotes the position of the passing pooling window. The average of all positions within the Y_F is calculated as a partition threshold θ, and the background feature map Y_B consists of parts smaller than θ. The formulas of θ and Y_B are denoted as

$$\theta = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} (Y_F)_{i,j} \quad (4)$$

$$Y_B = \begin{cases} 0, & Y_F > \theta \\ Y_F, & Y_F \le \theta. \end{cases} \quad (5)$$

The enhanced foreground feature map is obtained from the disparity between the above two components, expressed as

$$Y^* = (Y_F - Y_B). \quad (6)$$

The final output Z is shown below, where α represents the hyper parameter

$$Z = (1 - \alpha)X + \alpha Y^*. \quad (7)$$

The FEM-based network emphasizes the foreground areas containing target entities. By enhancing the weight of target areas and reducing unnecessary background information, the final prototype can encompass more distinct and accurate features. In addition, the CDCI basic network framework adopts dynamic extension, sharing parameters of the first three convolution blocks of ResNet-18 in the incremental stage and expanding the last convolution block. By applying FEM to the parameter sharing module, the model achieves more precise identification of information from previous categories. This approach prevents the overfitting that might arise from an excessive focus on new category features, ensuring that the model retains knowledge of existing categories while incorporating new categories.

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D. Domain Adaptive Module

The success of existing CI learning heavily relies on source and target domains coming from the same distribution. However, for remote sensing images, distribution shifts exist across different datasets due to variations in sensor types, imaging conditions, and capture angles. These issues limit the model's capability to generalize and apply to new data samples. Many existing approaches tackle the issue by adding adaptive layers next to the primary convolutional neural network (CNN), mainly using the maximum mean discrepancy (MMD) loss function to adjust the differences in marginal distributions. However, they tend to overlook the specific impact of different samples during the feature alignment process. To more effectively address these challenges, a novel dynamic alignment-guided multiscale DAM is proposed. This method optimizes the model's capacity to align data distributions across different domains by assigning dynamic weights to different samples, thereby improving the model's generalization prowess. The module is shown in Fig. 5.

1) Dynamic Alignment Structure: In traditional DA methods, the model weights are static for all samples and are not adjusted for different samples. In practical scenarios, various degrees of distribution shifts may form between different samples. Based on this, a DAS structure is proposed to realize the dynamic alignment of the network, expressed as follows:

$$W(x) = W_s(x) + W_d(x) \tag{8}$$

where $W(x)$ signifies the comprehensive dynamic weight matrix, $W_s(x)$ denotes the static convolutional kernel matrix, and $W_d(x)$ stands for the dynamic adjustment matrix. The squeeze-and-excitation (SE) module is used to emphasize the target features and suppress the useless features. Through the SE module, the dynamic scale matrix τ is obtained and represented as follows:

$$\tau = \phi(W_2\delta(W_1\text{AvgPool}(x))) \tag{9}$$

where W_1 and W_2 stand for the linear layers, δ represents the ReLU function, and ϕ refers to the softmax function. Specifically, the SE module first performs adaptive average pooling on the input sample, then calculates the dynamic scale matrix τ through a combination of two linear layers and a softmax layer. During this process, the parameters of the linear layers and softmax layer are learned and adjusted according to the feature representation of the input sample, which generates four different weight values τ_i suitable for the current sample. Each weight value corresponds to a different feature subspace, which can help the networks to better understand and utilize the different features in the input sample.

Then, we initialize four 1×1 convolutions, each with independent parameters that can learn different features. By utilizing them, we can obtain static matrices $\omega_i(x), i \in \{1, \ldots, 4\}$. $W_d(x)$ is expressed as follows:

$$W_d(x) = \sum_{i=1}^{4} \omega_i(x) \cdot \tau_i. \tag{10}$$

In the DAS structure, the static matrix $\omega_i(x)$ is described as the basis of the 1×1 convolutional weight space, and the dynamic proportional matrix τ is interpreted as the projection of the adjusted matrix in the corresponding weight subspace. Since the weight value τ_i is generated dynamically according to different input samples, the networks can adjust the contribution of the static matrix $\omega_i(x)$ according to these weights to select different feature subspaces. By combining both static and dynamic elements, the dynamic adjustment of the model weights can be realized, so that the model can select and emphasize the feature subspace most suitable for the current input sample, and better realize the dynamic alignment of different domains.

2) Multiscale Fusion Module: In dealing with complex RSSC tasks, traditional single-scale feature extraction methods often fail to capture detailed textures and subtle variations, resulting in the loss of critical information. However, the traditional CNN usually increases the network depth by the superposition of convolution layers, which can obtain more representative deep semantic features, but also ignore some details. To address the above problems, the inception multiscale module is introduced, which aims to extract information at different levels and scales from the input image in parallel, so as to obtain a more comprehensive and richer feature representation. The final output formula is expressed as follows:

$$F = \text{GAP}(F_1 \oplus F_2 \oplus F_3 \oplus F_4) \tag{11}$$

where F_i represents the features obtained from the different branches, GAP denotes global average pooling, and $\oplus$ is the element-wise sum.

Specifically, the image features passing through the shared module are first sent into multiple different convolutions (such as 1×1, 1×3, and 3×3) and pooling paths. Then, the outputs of different paths are fused at multiple scales. In this way, the feature vector of each position contains information about that position and information from different surrounding regions, increasing the ability of the network to perceive features across various scales. Moreover, the multiscale module can reduce the distribution shifts between the two domains on each scale, thereby enhancing the conditional invariance of the extracted features. The model's total loss comprises two components, defined as

$$L_{\text{total}} = L_{\text{cls}} + \lambda L_{\text{mmd}} \tag{12}$$

where L_{cls} denotes the cross-entropy loss derived from the labeled samples in the source domain. L_{mmd} represents the adaptation loss obtained from both domains, λ signifies the tradeoff parameter. As the number of iterations increases, the value of λ gradually transitions from 0 to 1. The pseudo-code flow of the proposed CDCI-RSSC is seen in Algorithm 1.

IV. Experiment

A. Datasets

It is worth noting that since the proposed approach is the first to implement CDCI in RSSC, there is no standard benchmark dataset for CDCI-RSSC. In this article, to verify the performance of the proposed CDCI framework, three existing remote scene classification datasets of varying scales are selected to build three different CD scenarios. Here is

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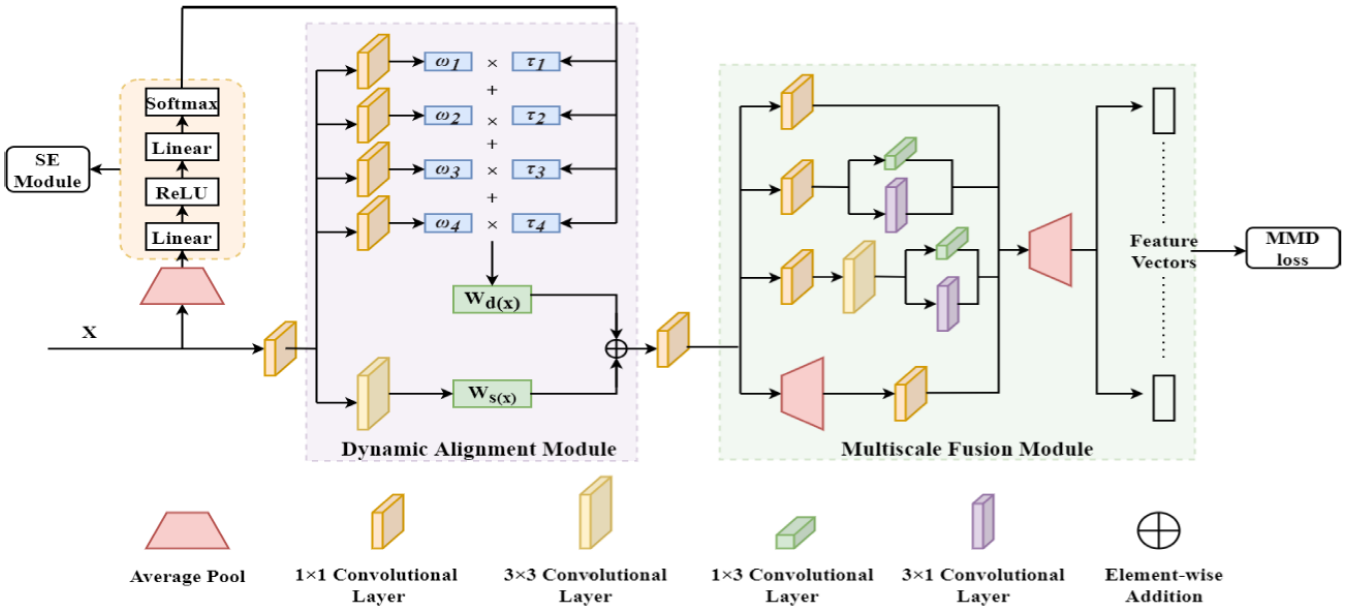


Fig. 5. Structure of the DAM. First, the dynamic alignment structure (DAS) calculates each sample's dynamic weight, and then the multiscale fusion module (MFM) completes feature extraction from different scales to better achieve distribution alignment.

Algorithm 1 CD CI for RSSC. L_{cls} Is the Cross-Entropy Loss Function, L_{mmd} Represents the Loss of DAM

Input: Base source domain dataset D^S, base target domain dataset. D^T_{old}, and incremental datasets $\{D^{T1}_{\text{train}}, D^{T2}_{\text{train}}, \ldots, D^{Ti}_{\text{train}}\}$

Output: The predicted labels on the testing sets D^T_{old} and $\{D^{T1}_{\text{test}}, D^{T2}_{\text{test}}, \ldots, D^{Ti}_{\text{test}}\}$

Step 1: base session
Train the model on the base training set D^S, which contains sufficient samples for each category
Get the feature Z corresponding to the FEM via Eq. (3)–Eq. (7)
Get the dynamic weight matrix $W(x)$ via Eq. (8)–Eq. (10) and the multiscale fusion feature F via Eq. (12) to better align the distribution shifts between different domains
Calculate loss function L_{cls} and L_{mmd}
Optimize the network parameters via Eq. (12)
Test the model on D^T_{old}
Step 2: incremental session
for *i=1 to n* **do**
    Train the model on the incremental training set D^{Ti}_{train}
    Learn new class features through the modular sharing-based dynamic extension method and calculate the loss function $L(x, y)$ via Eq. (2)
    Test the model on D^{Ti}_{test} and D^T_{old}
end

a detailed description of the three existing datasets and the process of building CD scenarios.

NWPU-RESISC45 is a dataset of RS images tailored for scene classification and recognition. This dataset encompasses 45 diverse scene categories, totaling 35 000 images. Each category contains 700 true-color RS images with dimensions of 256 × 256 × 3. The UCMerced-LandUse dataset, developed by the research team at the University of California, Merced, is an open-source dataset intended for land use classification and image analysis. It features 21 different categories with 2100 high-resolution aerial photographs, each with a resolution of 256 × 256 pixels. The Aerial Image Dataset (AID) is a comprehensive aviation dataset constructed using Google Earth images. It includes 30 aerial scene categories, each containing roughly 220 to 420 images, all sized at 600 × 600 pixels.

In this research, the previously mentioned datasets are utilized, with them being paired as the source and target domains, respectively. We choose partially overlapping categories as base classes and conduct CD testing while completing the continuous learning of the unknown class in the target domain. Taking AID → UCM as an example, we selected repeated nine classes from two datasets as the base class and tested the AID data as the training set (source domain), and UCM data as the test set (target domain). The incremental part consists of the remaining 12 classes, which are divided into 3 sessions, with each session containing four classes. Considering the possible difficulty in obtaining samples in the actual target domain, the training set is assigned to the test set in a 2:8 ratio in the incremental part, aiming to achieve better learning performances even with limited training samples.

B. Experimental Setting

Throughout the experiment, we not only focus on the CD testing performance during the base class process but also need to focus on the model's ability to learn novel classes. Meanwhile, a robust backbone network needs to be trained with sufficient data in the base classes to enable effective adaptation to new classes. Therefore, we construct three scenarios where the source domain data far exceeds the target data: AID →

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TABLE I
EXPERIMENT COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE AID → UCM DATASET CDCI SCENARIO

Type	Method	Session 0	Session 1	Session 2	Session 3	Average	Relative Improvement
Cross-domain	AMRAN (TGRS 2021) [16]	89.67	66.15	52.40	43.39	62.90	+12.38
	DSAN (TNNLS 2021) [35]	88.44	65.25	51.69	42.80	62.05	+13.23
	DATSNET (TGRS 2022) [30]	92.67	68.36	54.16	44.81	65.00	+10.28
	MEDM (TNNLS 2023) [36]	85.56	63.15	50.00	41.40	60.03	+15.25
	LDP-Net (CVPR 2023) [37]	29.29	26.73	19.41	15.89	22.83	+52.45
	SCSN (ISPRS 2023) [38]	83.56	61.70	48.90	40.48	58.66	+16.62
Class-incremental	BIC (CVPR 2019) [39]	74.56	64.59	65.00	61.24	66.35	+8.93
	DER (CVPR 2021) [34]	74.56	62.54	61.43	59.30	64.46	+10.82
	FOSTER (ECCV 2022) [40]	75.00	66.39	64.55	63.49	67.36	+7.92
	FETRIL (WACV 2023) [41]	74.56	61.86	62.86	58.98	64.57	+10.71
	DRC-CIL (ICCV 2023) [42]	77.78	16.48	13.90	15.38	30.86	+44.42
	MEMO (ICLR 2023) [43]	72.78	71.64	70.45	68.43	70.83	+4.45
CDCI	**CDCI-RSSC (ours)**	76.33	77.05	74.68	73.06	75.28	

TABLE II
EXPERIMENT COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE NWPU → UCM DATASET CDCI SCENARIO

Type	Method	Session 0	Session 1	Session 2	Session 3	Average	Relative Improvement
Cross-domain	AMRAN (TGRS 2021) [16]	93.22	68.77	54.48	45.11	65.40	+15.85
	DSAN (TNNLS 2021) [35]	92.44	68.20	54.03	44.73	64.85	+16.40
	DATSNET (TGRS 2022) [30]	88.15	65.00	51.49	42.63	61.82	+19.43
	MEDM (TNNLS 2023) [36]	91.78	67.70	53.64	44.19	64.33	+16.92
	LDP-Net (CVPR 2023) [37]	25.00	25.29	18.53	17.78	21.65	+59.60
	SCSN (ISPRS 2023) [38]	87.78	64.75	51.30	42.47	61.58	+19.67
Class-incremental	BIC (CVPR 2019) [39]	85.78	77.95	68.18	68.12	75.01	+6.24
	DER (CVPR 2021) [34]	84.78	79.84	70.19	70.59	76.35	+4.90
	FOSTER (ECCV 2022) [40]	84.11	76.39	67.27	63.76	72.88	+8.37
	FETRIL (WACV 2023) [41]	85.33	81.31	70.78	67.15	76.14	+5.11
	DRC-CIL (ICCV 2023) [42]	84.78	14.92	19.35	21.51	35.14	+46.11
	MEMO (ICLR 2023) [43]	84.89	80.74	75.39	75.05	79.02	+2.23
CDCI	**CDCI-RSSC (ours)**	87.67	83.44	76.56	77.31	81.25	

UCM, NWPU → AID, and NWPU → UCM, to conduct the experiment. Here, the arrow symbol (→) indicates the direction from the source domain to the target domain.

In this article, we adopt the widely recognized CIL framework memo as the baseline and select ResNet-18 as the backbone of the feature extractor. The algorithm is developed based on the PyTorch and Tesla V100 GPU. The model training is optimized using the SGD, with a batch size of 128, an initial training stage comprising 200 epochs, and a learning rate of 0.01. The epoch is set to 170 in the incremental learning phase, and the learning rate is adjusted to 0.1. Other parameters of the model are flexibly adjusted according to the specific dataset characteristics and task requirements. Furthermore, in this research, the automated data augmentation technique introduced in the FACT is employed to preprocess the training data. After each round, model performance is evaluated by test set D_i, and TOP-1 accuracy is recorded for analysis.

C. Comparison With Existing Methods

Most current CI learning methods are primarily applied to traditional public datasets, with relatively less application in RSSC. To better evaluate the proposed CDCI-RSSC framework, we compare it with some advanced CI learning architectures, replacing the original datasets according to the same experimental design. The methods compared include: BIC [39], DER [34], FOSTER [40], FETRIL [41], DRC-CIL [42], and MEMO [43]. Furthermore, to further demonstrate the superiority of the proposed method in new class learning, we select also six advanced CD learning methods for comparison, including AMRAN [16], DSAN [35], DATSNET [30], MEDM [36], LDP-Net [37], and SCSN [38]. Throughout the entire experiment, all comparative methods are implemented according to our CDCI architecture to ensure the fairness and effectiveness of the experimental results. We introduce the average accuracy Average $= (1/N)\sum_i A_i$ across all sessions to comprehensively assess the domain adaptability and new class learning capabilities of the model.

The experimental results in the AID → UCM dataset CDCI scenario are shown in Table I, and they are divided into two parts, the comparison of CD learning and CI learning methods. In the CD learning section, we do not focus on the accuracy of the first session since our approach does not entirely focus on domain adaption. We evaluate the continuous learning capability for new classes of each method through their average accuracy. Compared to the baseline method SCSN, our approach achieves a 16.62% performance improvement. In particular, LDP-Net operates as a cross-domain few-shot learning (CD-FSL) method, and its performance in the CDCI setting appears limited for the following reasons.

1) The CD-FSL model relies on a few samples to fine-tune the model to accommodate the new category. In the CDCI scenario, no further training on the existing categories in the source domain is conducted, resulting in poor performance in session 0.

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TABLE III
EXPERIMENT COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE NWPU → AID DATASET CDCI SCENARIO

Type	Method	Session 0	Session 1	Session 2	Session 3	Average	Relative Improvement
Cross-domain	AMRAN (TGRS 2021) [16]	93.13	67.86	52.02	42.91	63.98	+18.43
	DSAN (TNNLS 2021) [35]	92.02	67.05	51.40	42.40	63.22	+19.19
	DATSNET (TGRS 2022) [30]	97.17	70.80	54.14	44.77	66.72	+15.69
	MEDM (TNNLS 2023) [36]	92.85	67.66	51.87	42.78	63.79	+18.62
	LDP-Net (CVPR 2023) [37]	15.71	19.03	18.56	13.23	16.63	+65.78
	SCSN (ISPRS 2023) [38]	94.14	68.59	52.59	43.37	64.67	+17.74
Class-incremental	BIC (CVPR 2019) [39]	82.11	78.36	69.29	69.03	74.70	+7.71
	DER (CVPR 2021) [34]	85.48	78.91	69.12	65.89	74.85	+7.56
	FOSTER (ECCV 2022) [40]	85.55	77.70	68.26	62.09	73.40	+9.01
	FETRIL (WACV 2023) [41]	89.23	82.84	76.99	71.39	80.11	+2.30
	DRC-CIL (ICCV 2023) [42]	90.93	60.53	59.09	58.44	67.23	+15.18
	MEMO (ICLR 2023) [43]	84.19	81.82	79.01	75.87	80.22	+2.19
CDCI	**CDCI-RSSC (ours)**	86.53	84.27	80.77	78.05	82.41	

2) The model lacks a mechanism to retain old class knowledge, limiting its predictions to current task categories in each incremental learning phase, which affects long-term accuracy.

In the CI learning part, we adopt a similar baseline structure, focusing on the accuracy of each session as well as the overall average accuracy. Our method performs well in all sessions and has a lower forgetting rate. Compared to the baseline method MEMO, our approach shows a 4.45% improvement in performance. The experimental results of NWPU → UCM and NWPU → AID dataset CDCI scenarios are shown in Tables II and III, which achieves 2.23% and 2.19% improvement over the MEMO method. In summary, our method addresses both the limitations of CD methods for continuously learning new classes and overcomes the challenges of CI methods in facing differences in data distribution. In addition, our experiments across three different CD scenarios further demonstrate that the CDCI-RSSC method maintains broad applicability, even when dealing with disparities in the sizes of source and target domain datasets.

D. Ablation Studies

In this section, various ablation studies are performed to evaluate the efficacy of the proposed FEM and DAM. During the experiments, we select the AID → UCM, NWPU → UCM, and NWPU → AID datasets as CDCI scenarios, and MEMO as the framework structure. The ablation study is divided into three primary sections, with the experimental results comparison presented in Figs. 6–8.

1) MEMO + FEM: The FEM is designed into the baseline model to increase the weight of key regions and reduce background interference. In both scenarios, the module accuracy and results are improved by 2.78%, 1.59%, and 1.58% on the baseline structure MEMO, respectively.

2) MEMO + DAM: The DAM is integrated into the baseline structure, combining a dynamic alignment strategy and multiscale approach to achieve distribution alignment between the source and target domains. In both scenarios, the accuracy and results of this module are improved by 2.00%, 1.80%, and 1.02% over MEMO, respectively.

3) MEMO + FEM + DAM: We combine the DAM and FEM to enable the model to solve the distribution shifts between the different domains while focusing on the information of the key regions. It can be seen that in the two scenarios, our method is increased by 4.45%, 2.23%, and 2.19% over the MEMO, 1.67%, 0.64%, and 0.61% over MEMO + FEM, and 2.45%, 0.43%, and 1.17% over MEMO + DAM, respectively.

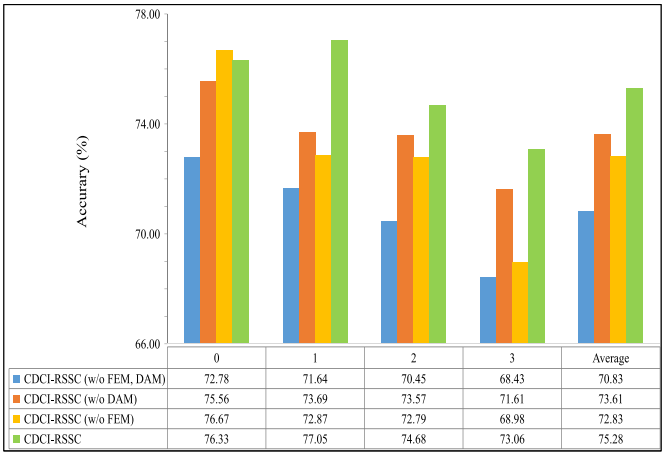


Fig. 6. Ablation study on the AID → UCM dataset CDCI scenario. w/o denotes without a certain module.

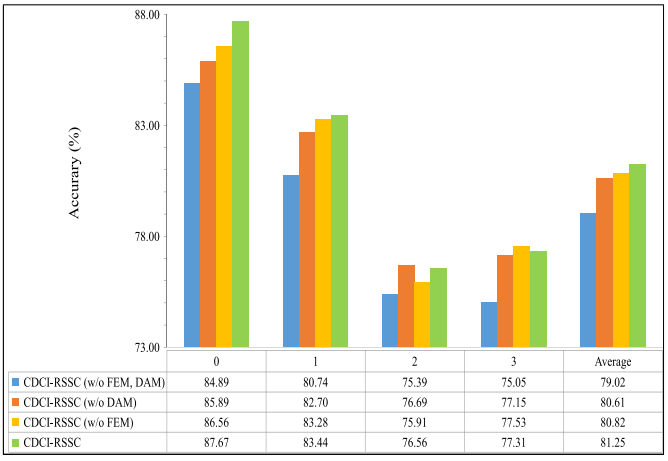


Fig. 7. Ablation study on the NWPU → UCM dataset CDCI scenario. w/o denotes without a certain module.

E. Parameters Sensitivity

1) Tradeoff Parameter λ: In this section, we analyze the effect of λ in the total loss function for the classification

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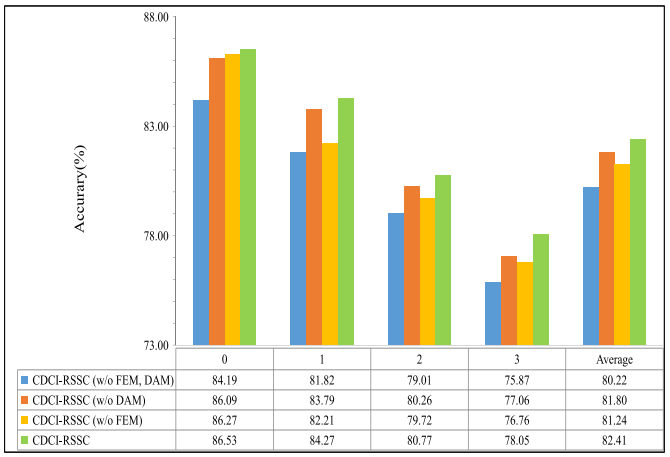


Fig. 8. Ablation study on the NWPU → AID dataset CDCI scenario. w/o denotes without a certain module.

TABLE IV
SENSITIVITY ANALYSIS OF THE TRADEOFF PARAMETER λ VALUES ON THE AID → UCM DATASET CDCI SCENARIO

λ	Session1	Session2	Session3	Session4	Average
0	72.78	71.64	70.45	68.43	70.83
0.3	73.33	71.31	72.99	70.00	71.91
0.6	75.44	74.18	73.70	71.08	73.60
0.9	74.89	71.97	74.29	72.10	73.31
Auto-balance	**76.33**	**77.05**	**74.68**	**73.06**	**75.28**

TABLE V
EXPERIMENT COMPARISON OF THE RUNNING TIMES AND ACCURACY OF DIFFERENT METHODS ON THE AID → UCM DATASET CDCI SCENARIO

Method	AMRAN	DER	FOSTER	FETRIL	DRC-CIL	CDCI-RSSC
Time (s)	2.57	1.44	2.03	1.46	**1.33**	1.51
Accuracy (%)	62.90	64.46	67.36	64.57	30.86	**75.28**

performance of the AID → UCM dataset CDCI scenario. λ is an automatically learned hyperparameter that is defined by the following formula:

$$\lambda = \frac{2}{1+\exp\left(-\frac{10\times\text{epoch}}{200}\right)} - 1. \tag{13}$$

As the number of epochs increases from 0 to 200, λ gradually changes from 0 to 1. This adjustment allows the model to concentrate on the classification loss in the early stages of training. As training progresses, the incrementing λ enhances the significance of the MMD loss term in the overall loss, thereby better addressing the distribution shifts between different domains. To demonstrate the effectiveness of λ automatically learning, we manually select several values of λ, respectively, 0, 0.3, 0.6, and 0.9. The results, as depicted in the accompanying Table IV, indicate that the automated learning hyperparameter λ produces significant positive results for each incremental session and mean accuracy.

2) Computational Cost: In this section, we analyze five methods for running time and average accuracy with our CDCI-RSSC method under the AID → UCM dataset CDCI scenario. According to the data shown in Table V, our method, compared to DER [34], FETRIL [41], and DRC-CIL [42], achieves a significant improvement in test accuracy despite a slight increase in runtime. Meanwhile, when compared to AMRAN [16] and FOSTER [40], our method shows clear advantages in both runtime and test accuracy. Overall, our method outperforms existing baselines, achieving higher accuracy and completing tasks within a reasonable time frame.

3) Base Classes: During the evaluation phase, the accuracy of the proposed CDCI-RSSC model across various scenarios is evidenced through the parameters adopted under the testing conditions. In this section, we focus on the specific effect of choosing the number of different classes during the base session. Therefore, we try different base categories, specifically 6, 9, and 12, while keeping the number of sessions in the incremental learning phase fixed. The experiments are performed in two different CDCI scenarios, NWPU → UCM and AID → UCM, DER and MEMO are used for comparisons. Table VI presents the average accuracy of each method under various test conditions. Since our model needs to consider CD factors when dealing with the base categories, the impact of different category numbers on classification performance is particularly significant, leading to notable differences in the results. Nonetheless, our method shows the best performance in various testing situations, further validating that our model can adapt to different testing conditions.

4) Incremental Sessions: In this section, we explore in depth the number of sessions in the incremental learning phase on experimental results. To perform this study, the number of categories in the base session is set as nine classes, and 2, 3, and 4 incremental sessions are selected for the experiment. Our experiments cover two different CD scenarios, NWPU → UCM and AID → UCM, comparative methods still using DER and MEMO as references. Table VII shows the average accuracy of each method under different test conditions. The results indicate that our model performs optimally across all conditions, and the test accuracy does not fluctuate significantly with the number of incremental sessions. This finding provides further evidence that our model has good generalization and the ability to consistently adapt to new categories.

F. t-SNE Visualization

In this section, t-SNE visualization analysis is used to validate the effectiveness of the CDCI-RSSC. We show the t-SNE feature embeddings of the session2 on the AID → UCM, NWPU → UCM, and NWPU → AID dataset CDCI scenarios. According to the principle of t-SNE, characteristics of samples belonging to the same category should cluster together. Figs. 9–11 demonstrate t-SNE visualization results under three comparative methods: the proposed CDCI-RSSC, the DER, and MEMO. The results reveal that although the DER and MEMO achieve some success, they suffer from large intraclass distances and easily confused interclass boundaries. In contrast, our t-SNE visualization exhibits clear classification boundaries and better intraclass cohesion.

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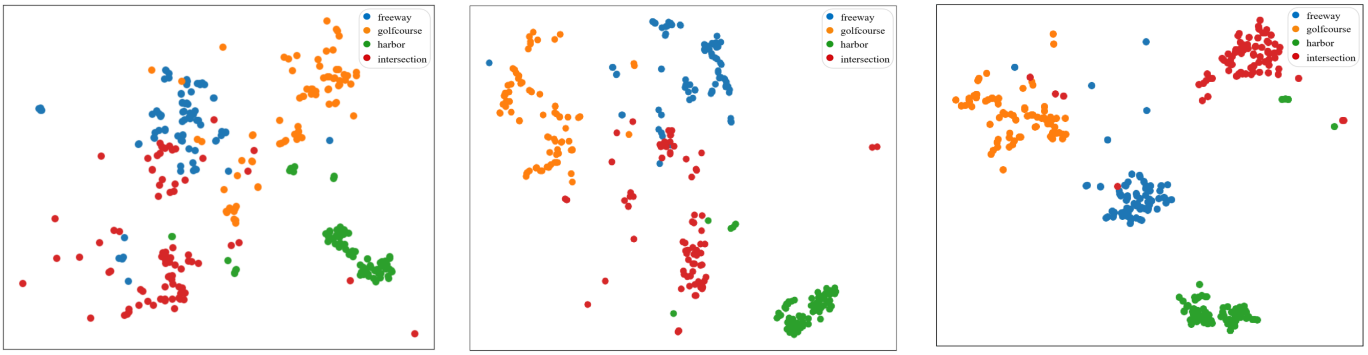


Fig. 9. t-SNE feature embeddings of the session2 on the AID → UCM dataset CDCI scenario, with four distinct colors representing different categories. (a) DER [34]. (b) MEMO [36]. (c) CDCI-RSSC (ours).

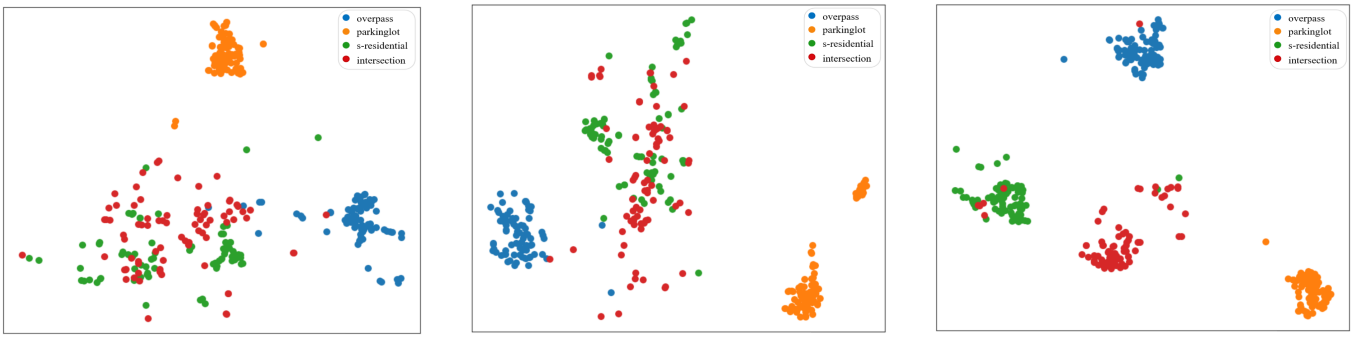


Fig. 10. t-SNE feature embeddings of the session2 on the NWPU → UCM dataset CDCI scenario, with four distinct colors representing different categories. (a) DER [34]. (b) MEMO [36]. (c) CDCI-RSSC (ours).

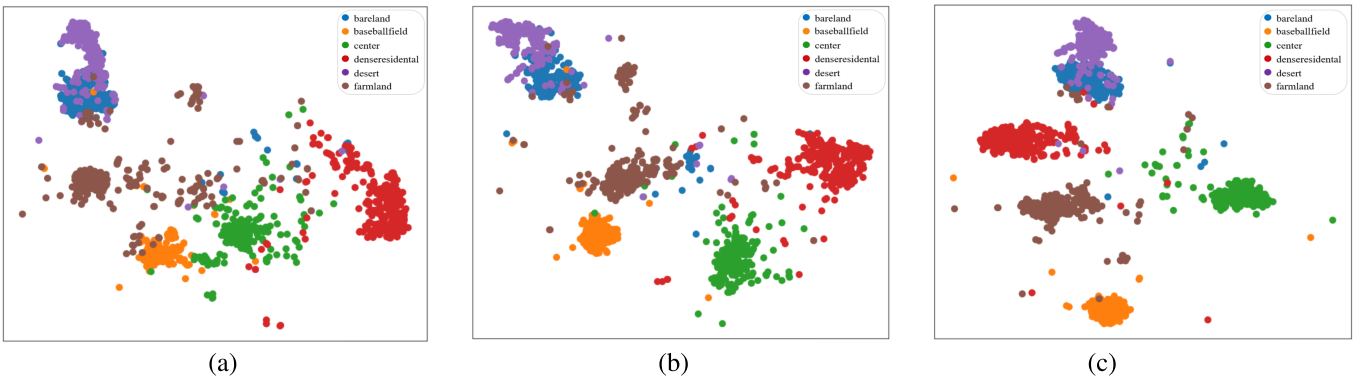


Fig. 11. t-SNE feature embeddings of the session2 on the NWPU → AID dataset CDCI scenario, with six distinct colors representing different categories. (a) DER [34]. (b) MEMO [36]. (c) CDCI-RSSC (ours).

TABLE VI

SENSITIVITY ANALYSIS OF THE BASE CLASS NUMBERS ON THE CDCI SCENARIO. BN DENOTES THE NUMBER OF BASE CLASSES. IK REPRESENTS THE NUMBER OF INCREMENTAL SESSIONS

Task Setup	NWPU→UCM			AID→UCM		
	DER [34]	MEMO [36]	**CDCI-RSSC (ours)**	DER [34]	MEMO [36]	**CDCI-RSSC (ours)**
B6-I3	74.78	77.46	**83.06**	67.57	64.48	**75.12**
B9-I3	76.35	79.02	**81.25**	64.46	70.83	**75.28**
B12-I3	78.50	80.86	**82.28**	65.86	69.25	**73.10**

These findings further confirm that our approach can effectively capture new class features with higher discriminative capability, thereby significantly enhancing classification performance.

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TABLE VII
SENSITIVITY ANALYSIS OF THE INCREMENTAL SESSION NUMBERS ON THE CDCI SCENARIO. BN DENOTES THE NUMBER OF BASE CLASSES. IK REPRESENTS THE NUMBER OF INCREMENTAL SESSIONS

Task	NWPU→UCM			AID→UCM		
Setup	DER [34]	MEMO [36]	**CDCI-RSSC (ours)**	DER [34]	MEMO [36]	**CDCI-RSSC (ours)**
B9-I2	76.94	78.74	**81.20**	66.87	69.45	**75.52**
B9-I3	76.35	79.02	**81.25**	64.46	70.83	**75.28**
B9-I4	76.65	79.15	**81.47**	65.78	69.73	**75.56**

V. CONCLUSION

In this article, we are the first to reveal the importance of CDCI in the field of remote sensing, which provides an innovative approach and perspective for RSSC. Our goal is to establish a CDCI-RSSC framework that not only addresses distribution shifts between source and target domains but also facilitates continuous learning of unknown categories. Specifically, we design a modular sharing-based dynamic extension approach to address the issue of memory consumption caused by traditional methods, propose a dynamic alignment guided multiscale DAM to effectively alleviate the distribution shifts between the source and target domains, build an FEM by increasing the weight of key areas to improve the representation of foreground.

As for CDCI-RSSC, it is still in its early stages, and many issues deserve further exploration. In the experiment, we find that the domain adaptive method applied to the existing class CI learning model does not achieve the desired effect. In future work, designing a high-precision identification module to automatically distinguish between known and unknown classes in the target domain, and building a CI learning framework to adapt different scenarios in the existing CD-RSSC models, are two key tasks we will focus on.

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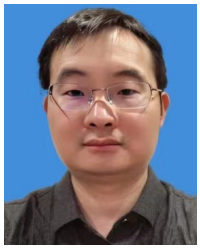
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